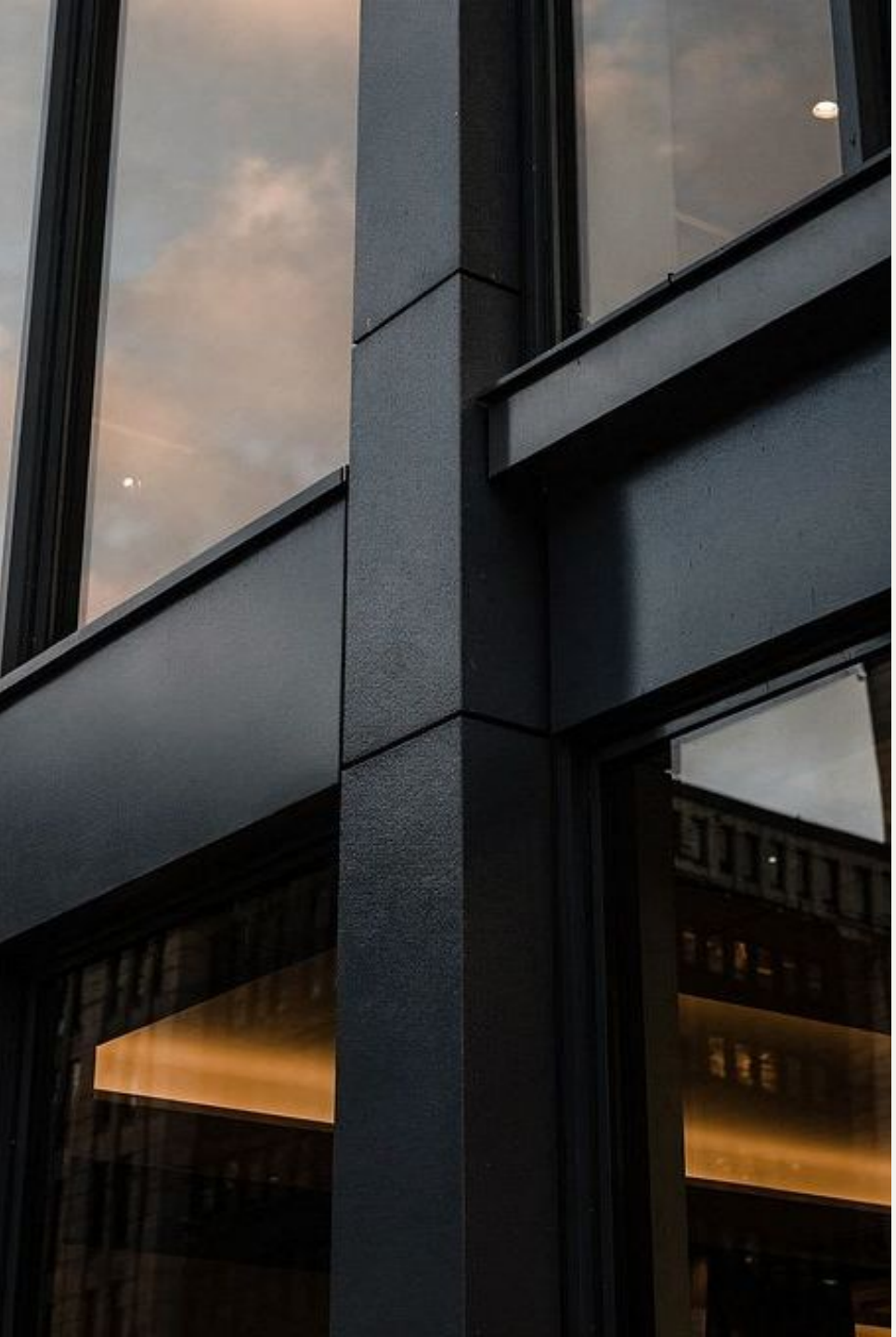


FitBuddy Project Report

PROJECT NAME: FitBuddy – AI Fitness Plan Generator using Gemini Models

TEAM ID:

A PROJECT REPORT

Submitted by

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In partial fulfilment for the award of the degree

of

BACHELOR

OF

COMPUTER SCIENCE

SRIDEVI ARTS AND SCIENCE COLLEGE KRISHNAPURAM, PONNERI-601 204

SEPTEMBER-2026



SMART INTERNZ × TN SKILL

FitBuddy – AI Fitness Plan Generator using Gemini Models

FitBuddy is a web-based application that uses AI to generate personalized workout plans and nutrition tips based on a user's fitness goals, such as weight loss, muscle gain, or general wellness. Built with FastAPI and integrated with Google's Gemini AI models, FitBuddy simplifies the process of creating structured and effective fitness routines.

The application takes in user information—like age, weight, goal, and preferred workout intensity—and generates a customized 7-day fitness plan. Users can also submit feedback to refine their plan using AI. A complementary nutrition or recovery tip is also provided based on their selected goal.

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Institution

Sridevi Arts and Science College, Krishnapuram, Ponneri – September 2026

Key Scenarios & Technical Architecture

4	2	7
Scenarios	Gemini Models	Day Plan
Core user & admin use cases	Pro & Flash for AI generation	Personalized workout schedule

Four Core Scenarios

01

Personalized 7-Day Plan

User inputs details (age, weight, goal, intensity) → AI generates a structured day-by-day workout schedule.

02

Feedback-Based Update

User submits feedback (e.g., "more cardio") → AI regenerates an improved, dynamic plan.

03

Nutrition & Recovery Tips

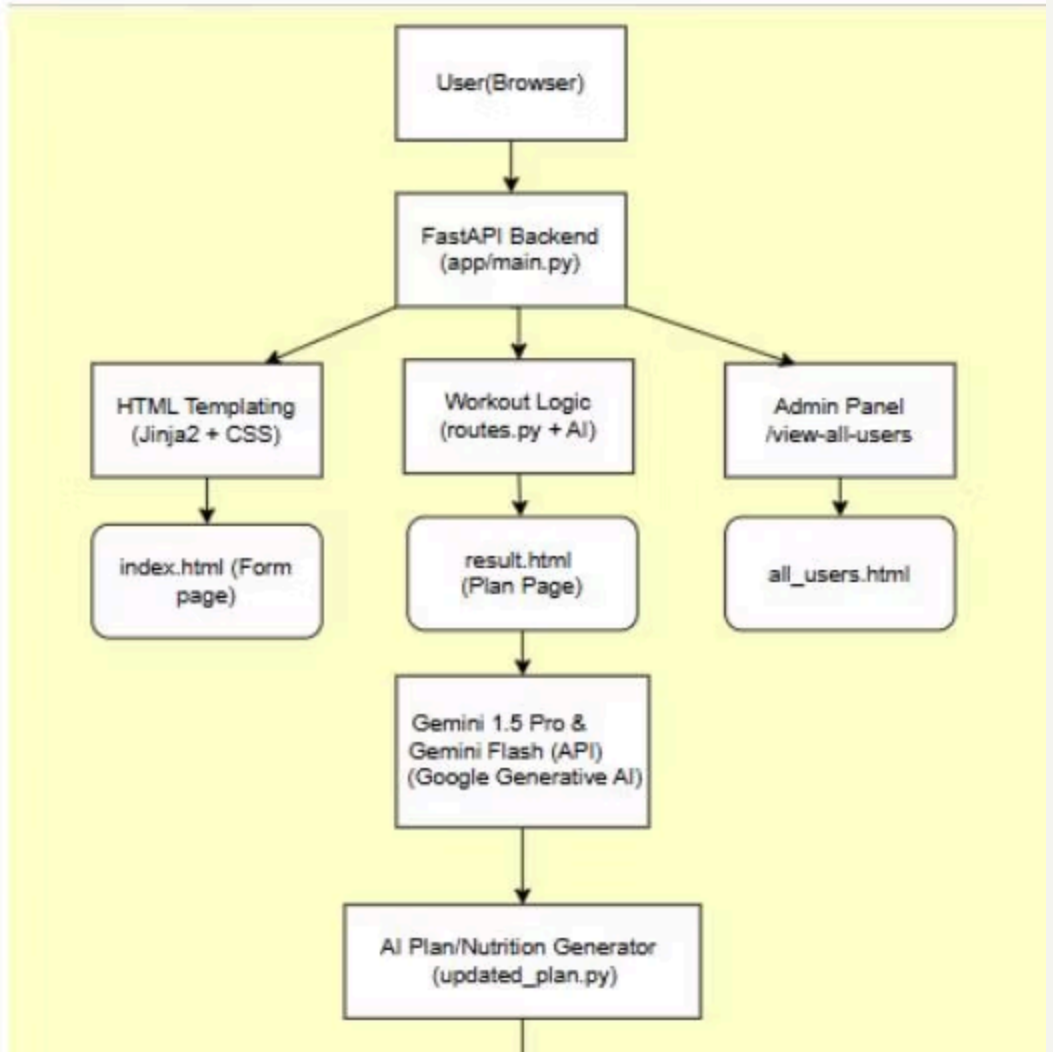
Goal-specific tips delivered via Gemini Flash (e.g., protein advice for muscle gain).

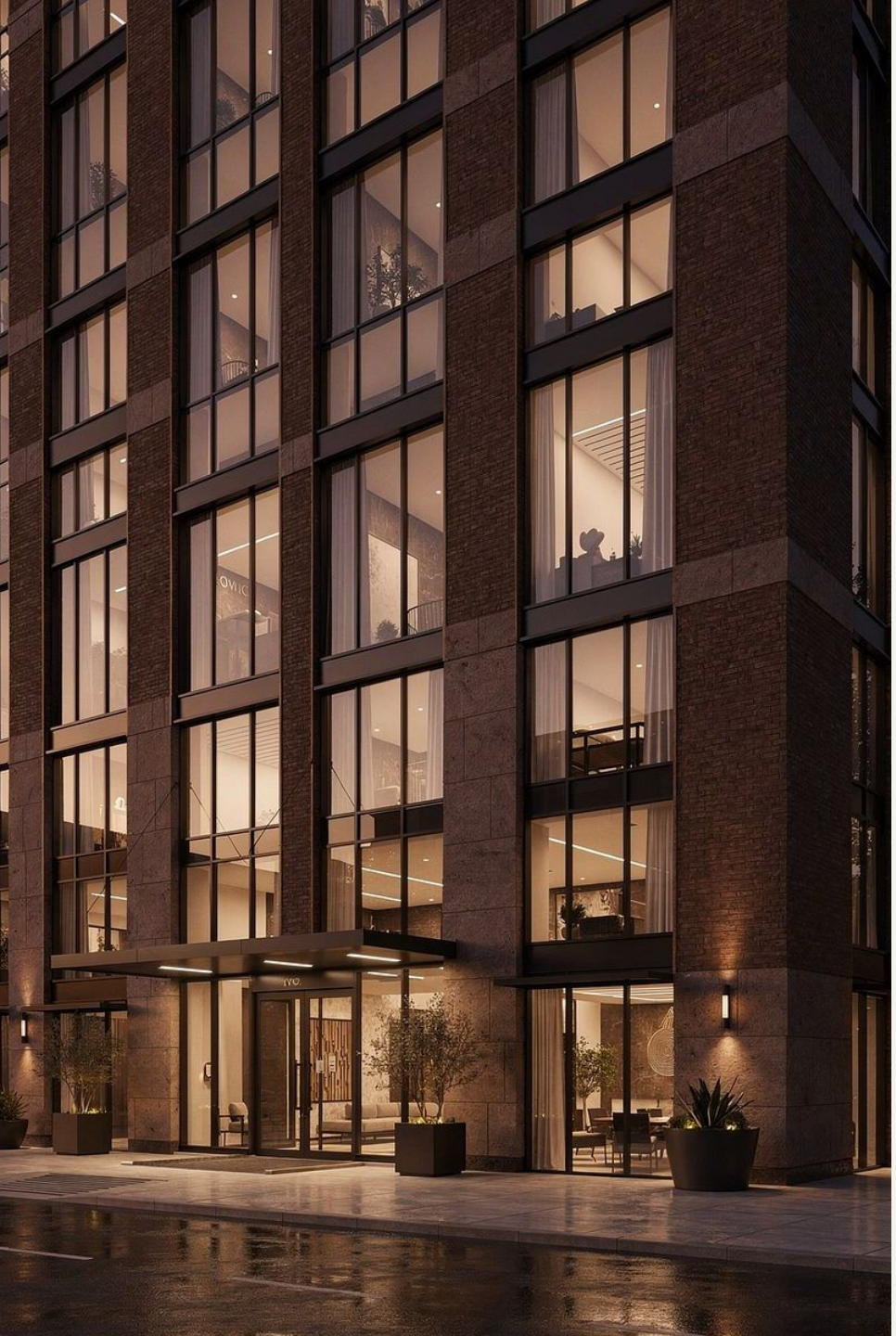
04

Admin Dashboard

Coach or admin views all users, original and updated plans in a structured table view.

Technical Architecture





Milestone 1: Model Selection & Architecture

Selecting the right generative AI models and defining the application structure.

1

Activity 1.1 – AI Model Research & Selection

Evaluated Hugging Face models (T5, BART, DistilBERT, Bloom) but found they required GPU hosting or fine-tuning. Selected **Gemini 1.5 Pro** for structured 7-day workout generation and **Gemini Flash** for fast, lightweight nutrition tips — both API-based, requiring minimal prompt tuning and natively understanding structured formatting.

2

Activity 1.2 – Application Architecture

FitBuddy uses a modular FastAPI architecture: **Frontend** via HTML + Jinja2 templates; **Backend** via FastAPI for routing; **AI Layer** using Gemini Pro & Flash APIs; **Database** using SQLite + SQLAlchemy ORM. Workout generation and feedback updates use Gemini Pro; nutrition tips use Gemini Flash.

3

Activity 1.3 – Development Environment Setup

Created a Python virtual environment, installed FastAPI, Uvicorn, Jinja2, SQLAlchemy, and Google Generative AI SDK. Project structure organized into `/app`, `/templates`, and `/static` folders. Server launched via `uvicorn app.main:app --reload` at `http://127.0.0.1:8000`.

Milestone 2: Core Functionalities Development

🏋️ Workout Plan Generation

File: `gemini_generator.py`

Sends user goal and intensity to **Gemini 1.5 Pro**, returning a structured 7-day plan with warm-up, main workout (sets & reps), and cooldown for each day. Rendered in `result.html` via `<pre>` blocks.

🥗 Nutrition Tip Generation

File: `gemini_flash_generator.py`

Uses **Gemini Flash** to generate a fast, concise nutrition or recovery tip based on the user's fitness goal (e.g., protein advice for muscle gain). Displayed beneath the workout plan.

🔄 Feedback-Based Plan Update

File: `updated_plan.py`

Passes the original plan + user feedback to **Gemini 1.5 Pro**, which returns a revised plan. Updated plan is stored alongside the original for comparison.

🗄️ User & Plan Storage

File: `database.py`

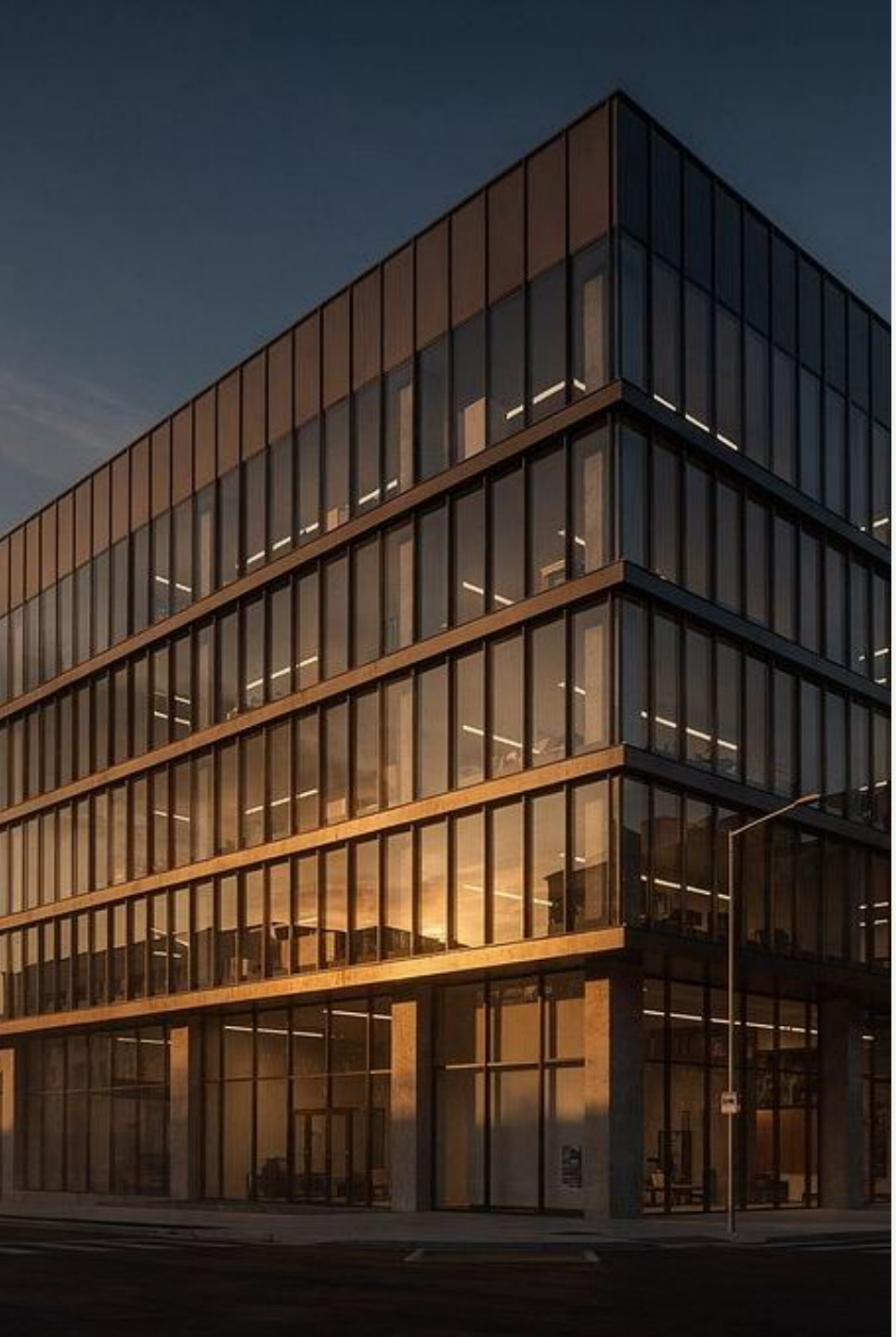
All user info (name, age, weight, goal, intensity) and AI-generated plans stored in **SQLite via SQLAlchemy ORM**. Functions: `save_user()`, `save_plan()`, `update_plan()`, `get_original_plan()`, `get_user()`.

Activity 2.2 – FastAPI Routing & Input Processing

- HTML form in `index.html` captures username, user_id, age, weight, goal, and intensity
- FastAPI captures inputs using `Form(...)` parameters
- Pydantic schemas (`UserInput`, `FeedbackRequest`) validate structure

API Route ↣ AI Function Mapping

- `/generate-workout` → `generate_workout_gemini()` + `generate_nutrition_tip_with_flash()`
- `/submit-feedback` → retrieves plan + calls `update_workout_plan()`
- `/view-all-users` → fetches all DB records for admin view



Milestone 3: routes.py Development

The routing layer bridging frontend templates, Gemini AI modules, and the SQLite database.

1 / – Home Route

Returns the homepage (`index.html`) with no form processing logic.

2 /generate-workout – Plan Generator

Receives user input via `Form(...)`, constructs a `UserInput` Pydantic model, calls `generate_workout_gemini()` via Gemini 1.5 Pro and `generate_nutrition_tip_with_flash()` via Gemini Flash, then stores user details and the plan before rendering `result.html`.

3 /submit-feedback – Feedback Plan Update

Captures feedback and `user_id`, retrieves original plan from DB, calls `update_workout_plan()` via Gemini 1.5 Pro, saves the revised plan, and renders the updated result on `result.html`.

4 /view-all-users – Admin View

Queries all users and their plans from the DB, passes data to `all_users.html`. Admin can view username, age, weight, goal, intensity, and both original and updated workout plans.

FRONTEND DEVELOPMENT

Milestone 4: UI Design & Jinja2 Templates

index.html – User Input Form

Main entry point capturing username, user_id, age, weight, fitness goal, and workout intensity. Uses a gym-themed background image, Google Fonts (Roboto), Flexbox layout, and submits to `/generate-workout`. Ensures all experience levels can get started quickly.

all_users.html – Admin Dashboard

Loops over all user records using Jinja2's `{% for user in users %}`. Displays user ID, name, age, weight, goal, intensity, and both original and updated workout plans in a structured table. Enables trainers and admins to monitor all client progress in one view.

result.html – Personalized Plan Page

Displays the 7-day AI-generated workout plan using `{{ workout_plan }}`, a nutrition tip via `{{ nutrition_tip }}`, and user summary (`{{ username }}`, `{{ goal }}`, `{{ intensity }}`, etc.). Also includes the feedback form submitted via POST to `/submit-feedback`.

CSS & Responsive Design

Embedded CSS with Flexbox centers content vertically and horizontally across all pages. Media queries ensure mobile responsiveness. Buttons feature hover effects; input fields and result blocks have padding, shadows, and rounded corners for a modern, fitness-oriented aesthetic.

Milestone 5: Deployment & Testing

8000

Local Port

5

Key Files

3

HTML Pages

1

DB File

Virtual Environment Setup

Create and activate a virtual environment, then install dependencies via `pip install -r requirements.txt` (FastAPI, Uvicorn, SQLAlchemy, Google Generative AI SDK).

Configure Environment Variables

Store the Gemini API key securely using `GOOGLE_API_KEY=your_key_here` in a `.env` file loaded via `python-dotenv`.

Launch & Access Locally

Start the server with `uvicorn app.main:app --reload`. Access the app at `http://127.0.0.1:8000` and interactive API docs at `http://127.0.0.1:8000/docs`.

Testing & Verification

Verify all routes: home form, plan generation, feedback submission, and admin dashboard. Confirm AI responses render correctly and SQLite records persist between sessions.



Application Screens

Key pages of the FitBuddy web interface demonstrating the end-to-end user journey.



Home Page

Input form for name, user ID, age, weight, fitness goal, and workout intensity. Click **"Generate Plan"** to trigger AI plan generation.



Workout Plan Page

Displays user summary and the **7-day AI-generated workout plan** with daily warm-up, main exercises (sets & reps), and cooldown guidance.



Feedback Page

User enters their unique ID and feedback (e.g., "more cardio"). The system **regenerates an updated plan** via Gemini Pro and confirms with a success message.

4

Admin Dashboard

View all users and their original and updated workout plans in a structured table. Enables coaches to monitor progress and compare plan versions across multiple clients.

CONCLUSION

FitBuddy – A Scalable AI-Powered Fitness Solution

Gemini AI Core

Gemini 1.5 Pro for structured workout generation & feedback updates; Gemini Flash for fast nutrition tips

FastAPI Backend

Modular routing with Pydantic validation, Jinja2 templating, and SQLAlchemy ORM for full-stack integration

Dynamic Feedback Loop

Users refine their plans through feedback, keeping routines adaptive and goal-aligned over time

Admin Oversight

Dedicated dashboard for coaches and administrators to track all user plans, original and updated versions

Scalable & Adaptable

Local SQLite deployment ready for cloud migration; architecture supports multi-user fitness institutions

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FitBuddy demonstrates how cutting-edge generative AI can be integrated into everyday wellness routines to promote healthy, consistent, and goal-oriented lifestyles. It stands as a scalable solution that personalizes the fitness journey and evolves with the user's progress and preferences.